

BRAC UNIVERSITY

**Department of Electrical and Electronic
Engineering School of Engineering**

INTRODUCTION TO COMMUNICATION ENGINEERING

Lecture 01: Introduction

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[Date]

EEE341

1.1 COMMUNICATION SYSTEMS

Figure 1. 1 presents three typical communication systems:

- a wire-line telephone-cellular phone connection,
- a TV broadcasting system, and
- a wireless computer network.



Figure 1.1. Some examples of communications systems.

The most efficient and effective way to learn about communication is by studying the major functional blocks common to practically all communication systems. A **typical communication system model** is shown in Fig. 1.2. The key components of a communication system are as follows.

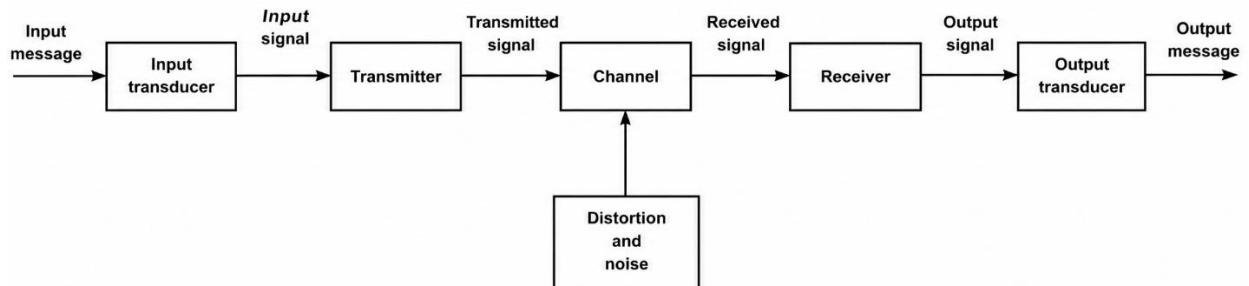


Figure 1.2. Communication system.

The **source** originates a message, such as a human voice, a television picture, an e-mail message, or data. If the data is nonelectric (e.g., human voice), it must be converted by an **input transducer** into an electric waveform referred to as the **baseband signal or message signal** through physical devices such as a microphone.

The **transmitter** modifies the baseband signal **for efficient transmission**. The transmitter may consist of one or more subsystems: an A/D converter, an encoder, and a modulator. Similarly, the receiver may consist of a demodulator, a decoder, and a D/A converter.

The **channel** is a medium of choice that can convey the electric signals at the transmitter output over a distance. A typical channel can be a pair of twisted copper wires (telephone and DSL), a coaxial cable (television and internet), an optical fiber, or a radio link.

The **receiver** reprocesses the signal received from the channel by reversing the signal modifications made at the transmitter and removing the distortions made by the channel. The receiver output is fed to the **output transducer**, which converts the electric signal to its original form—the message.

The **destination** is the unit to which the message is communicated.

A **channel** is a physical medium that behaves partly like a filter that generally **attenuates** the signal and **distorts** the transmitted waveforms.

- The **signal attenuation** increases with the length of the channel.
- **Signal waveforms are distorted** because of physical phenomena such as frequency-dependent gains, multipath effects, and Doppler shift.
- For example, a **frequency-selective** channel causes **different amounts of**

attenuation and phase shift to different signal frequency components.

In a practical environment, signals passing through communication channels **not only experience channel distortions but also are corrupted along the path by undesirable interferences and disturbances lumped under the broad term noise**. These interfering signals are random and unpredictable from external and internal sources.

External noise includes interference signals transmitted on nearby channels, **human-made noise** generated by faulty contact switches of electrical equipment, **automobile ignition** radiation, fluorescent lights or natural noise from lightning, **microwave ovens, and cellphone emissions**, as well as electric storms and solar and intergalactic radiation. With proper care in system design, external noise can be minimized or even eliminated in some cases.

Internal noise results from the **thermal motion of charged particles in conductors**, random emission, and diffusion or recombination of charged carriers in electronic devices. Proper care can reduce the effect of internal noise **but can never eliminate it**.

“Noise is one of the underlying factors that limit the rate of telecommunications.

- Thus, the channel distorts the signal in practical communication systems, and **noise accumulates along the path**.
- Worse yet, the signal strength decreases while the noise level remains steady regardless of the distance from the transmitter.
- Thus, the signal quality is continuously worsening along the length of the channel.
- These are the key challenges that we must face in designing modem communication systems.

1.2 ANALOG AND DIGITAL MESSAGES

Messages are digital or analog.

Digital messages are **ordered combinations of finite symbols or codewords**.

- For example, **printed English** consists of 26 letters, 10 numbers, a space, and several punctuation marks. Thus, a text document written in English is a digital message constructed from the ASCII keyboard **of 128 (2^7 >> 7 bits long codeword) symbols**.

- Similarly, a **Morse-coded telegraph message** is a digital message constructed from a set of only **two symbols-dash and dot**. It is therefore a **binary** message, implying **only two symbols**.
- A digital message constructed **with M symbols** is called an **M -ary message**.

Analog messages, on the other hand, are characterized by data **whose values vary over a continuous range and are defined for a continuous range of time**.

- For example, a **particular speech waveform** has **amplitudes that vary over a continuous range**. Over a given time interval, **an infinite number of possible different speech waveforms exist**, in contrast to only a finite number of possible digital messages.

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	.
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BELL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	;	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16	10	10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[END OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	100000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

Fig. ASCII table [3].

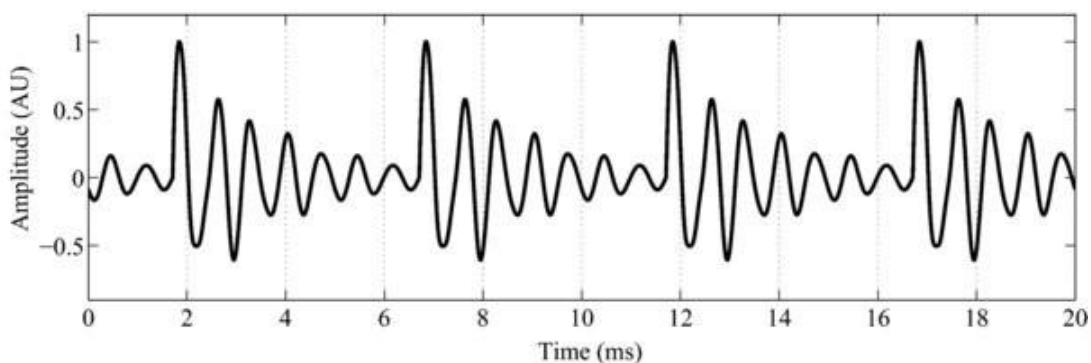


Fig. A speech waveform [4].

1.2.1 Why are digital technologies better?

The answer has to do with both **economics and quality**.

- The case for **economics** is made by noting the **ease of adopting** versatile, powerful, and inexpensive high-speed digital microprocessors.
- At the **quality level**, a **prominent feature** of digital communications is the enhanced immunity of digital signals to noise and interference.

❖ Digital messages are transmitted as a **finite set of electrical waveforms**.

For digital systems

- A digital message is generated from a **finite alphabet**, while each character in the alphabet can be represented by one waveform or a sequential combination of such waveforms.

For example, **in sending messages via Morse code**,

- **a dash** can be transmitted by an electrical pulse of amplitude $A/2$ and
- **a dot** can be transmitted by a pulse of negative amplitude $-A/2$ (Fig 1.3a).

In an **M -ary case**, **M distinct electrical pulses (or waveforms)** are used; each of the M pulses represents one of the M possible symbols.

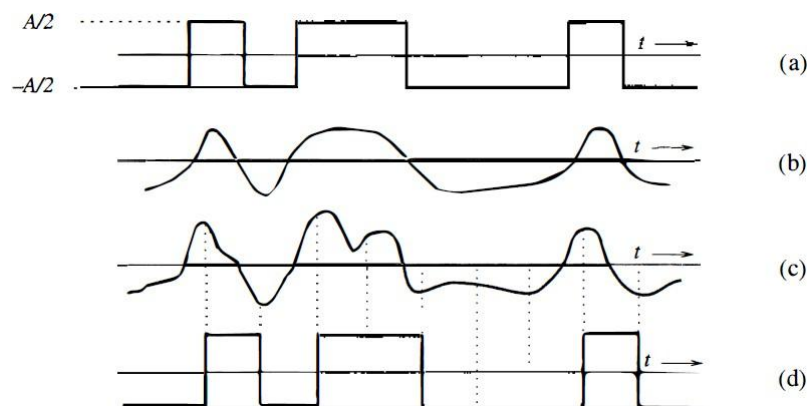


Figure 1.3. (a) Transmitted signal. (b) Received distorted signal (without noise). (c) Received distorted signal (with noise). (d) Regenerated signal (delayed).

- Once transmitted, the **receiver must extract the message** from **a distorted and noisy signal** at the channel output. **Message extraction is often easier from digital signals than from analog signals** because the **digital decision must belong to the finite-sized alphabet**.

- **Consider a binary case:** two symbols are encoded as rectangular pulses of amplitudes $A/2$ and $-A/2$.
- The **only decision at the receiver** is to select between **two possible pulses** received; the **fine details of the pulse shape are not an issue**.
- **A finite alphabet leads to noise and interference immunity**.

- The **receiver's decision** can be made **with reasonable certainty even if the pulses**

have suffered modest distortion and noise (Fig. 1.3).

- The digital message in Fig. 1.3a is distorted by the channel, as shown in Fig. 1.3b.
- Yet, if the distortion is not too large, we can recover the data without error because we need to make only a simple binary decision: *Is the received pulse positive or negative?*
- Figure 1.3c shows the same data with channel distortion and noise. Here again, the data can be recovered correctly as long as the distortion and the noise are within limits.

In contrast, for analog systems,

- the waveform shape itself in an analog message carries the needed information, and
- even a slight distortion or interference in the waveform will show up in the received signal.

Clearly, a digital communication system is more rugged than an analog communication system in the sense that it can better withstand noise and distortion (as long as they are within a limit).

❖ (1.2.2) Viability of distortionless regenerative repeaters

One main reason for the superior quality of digital systems over analog ones is the viability of regenerative repeaters and network nodes.

- Repeater stations are placed along the communication path of a digital system at distances short enough to ensure that noise and distortion remain within a limit. This allows pulse detection with high accuracy.
- At each repeater station, or network node, the incoming pulses are detected such that new, "clean" pulses are retransmitted to the next repeater station or node.
- This process prevents the accumulation of noise and distortion along the path by cleaning the pulses at regular repeater intervals.
- We can thus transmit messages over longer distances with greater accuracy.

For analog systems,

- Signals and noise within the same bandwidth cannot be separated.
- Repeaters in analog systems are basically filters plus amplifiers and are not "regenerative."
- Thus, it is impossible to avoid the in-band accumulation of noise and distortion along the path.
- As a result, the distortion and the noise interference can accumulate over the entire transmission path as a signal traverses through the network.
- To compound the problem, the signal is attenuated continuously over the transmission path.
- Thus, with increasing distance the signal becomes weaker, whereas the distortion and the noise accumulate more.

- Ultimately, the signal, overwhelmed by distortion and noise, is buried beyond recognition.

Amplification is of little help since it enhances both the signal and the noise equally. Consequently, the distance over which an analog message can be successfully received is limited by the first transmitter's power.

Despite these limitations, analog communication was used widely and successfully in the past for short-to medium-range communications.

Nowadays, because of the advent of optical fiber communications and the dramatic cost reduction achieved in the fabrication of high-speed digital circuitry and digital storage devices, almost all new communication systems being installed are digital.

But some old analog communication facilities are still in use, including those for AM and FM radio broadcasting.

1.2.3 Analog-to-Digital (A/D) Conversion

Despite the differences between analog and digital signals, a meeting ground exists between them: the conversion of analog signals to digital signals (A/D conversion).

A key device in electronics, the analog-to-digital (A/D) converter, enables digital communication systems to convey analog source signals such as audio and video.

- Generally, analog signals are continuous in time and in range; that is,
 - they have values at every time instant, and
 - their values can be anything within the range.
- On the other hand, digital signals exist
 - only at discrete points of time, and
 - they can take on only finite values.

A/D conversion can never be 100% accurate.

Since, however, human perception does not require infinite accuracy, A/D conversion can effectively capture necessary information from the analog source for digital signal transmission.

Two steps take place in A/D conversion:

1. a continuous-time signal is first sampled into a discrete-time signal, whose
2. continuous amplitude (signal level) is then quantized into a discrete-level signal.

Sampling

First, the frequency spectrum of a signal indicates the relative magnitudes of various frequency components.

The sampling theorem states that if the highest frequency in the signal spectrum is B (in hertz), the signal can be

reconstructed from its discrete samples, **taken uniformly at a rate not less than $2B$ samples per second.**

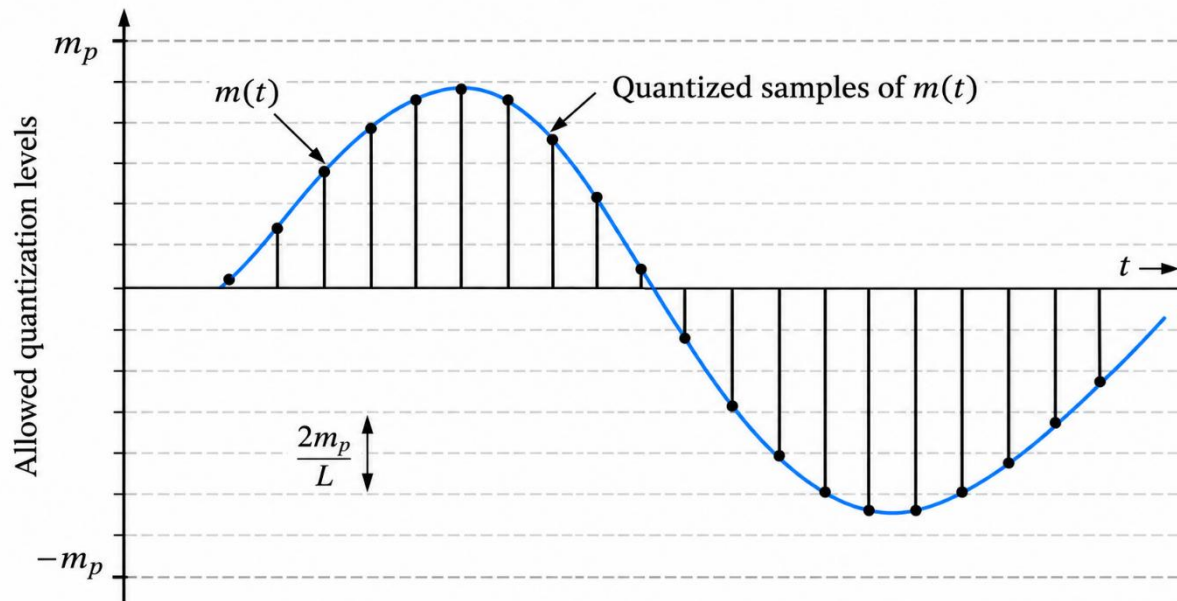


Figure 1.4. Analog-to-digital conversion of a signal.

This means that to **preserve the information from a continuous-time signal**, we need to **transmit only its samples (Fig. 1.4).**

However, the **sample values are still not digital because they lie in a continuous dynamic range.**

Quantization

Here, the **second step of quantization** comes to rescue. **Through quantization, each sample is approximated, or "rounded off," to the nearest quantized level**, as shown in Fig. 1.4.

- If amplitudes of the message signal $m(t)$ lie in the range $(-m_p, m_p)$, the quantizer **partitions the signal range into L intervals**.
- Each **sample amplitude is approximated by the midpoint** of the interval in which the sample value falls.
- Each **sample is now represented by one of the L numbers**.
- **The information is thus digitized.**

Hence, **after the two steps of sampling and quantizing**, the analog-to-digital (A/D) conversion is completed.

Note that the price for all these benefits of digital communication is paid in terms of increased

- **processing complexity and**
- **bandwidth of transmission.**

1.2.4 Pulse-Coded Modulation -A Digital Representation

Once the A/D conversion is over, the original analog message is represented by a sequence of samples, each of which takes on one of the L preset quantization levels.

- The transmission of this quantized sequence is the task of digital communication systems.
- For this reason, signal waveforms must be used to represent the quantized sample sequence in the transmission process.
- **Pulse-coded modulation (PCM)** is a very simple and yet common mechanism for this purpose. Note that, one information bit refers to one binary digit of 1 or 0.

The idea of PCM is to represent each quantized sample by an ordered combination of two basic pulses:

- $P_1(t)$ representing 1 and
 - $P_0(t)$ representing 0.
- Because each of the L possible sample values can be written as a bit string of length $\log_2 L$, each sample can therefore also be mapped into a short pulse sequence that represents the binary sequence of bits.
 - For example, if $L = 16$, then, each quantized level can be described uniquely by 4 bits. If we use two basic pulses, $P_1(t) = A/2$ and $P_0(t) = -A/2$. A sequence of four such pulses gives $2 \times 2 \times 2 \times 2 = 16$ distinct patterns, as shown in Fig.1.5.
 - We can assign one pattern to each of the 16 quantized values to be transmitted.
 - Each quantized sample is now coded into a sequence of four binary pulses.

This is the principle of PCM transmission, where signaling is carried out by means of only two basic pulses (or symbols).

The binary case is of great practical importance because of its simplicity and ease of detection. Much of today's digital communication is binary.

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

Figure 1.5. Example of PCM encoding.

1.3 CHANNEL EFFECT, SIGNAL-TO-NOISE RATIO, AND CAPACITY

In *designing communication systems*, it is important to **understand and analyze important factors** such as

- the **channel characteristics**,
- the **signal characteristics**,
- the **relative noise strength**,
- the **maximum number of bits that can be sent over a channel** per second, and,
- ultimately, the **signal quality**.

1.3.1 Signal Bandwidth and Power

In a given (digital) communication system, the **fundamental parameters and physical limitations** that **control the rate and quality** are the **channel bandwidth B** and the **signal power P_s** . if a channel can transmit with reasonable fidelity a signal whose frequency

The **bandwidth of a channel** is the range of frequencies that it can transmit with reasonable fidelity. For example,

components vary from 0 Hz (dc) up to a maximum of 5000 Hz (5 kHz), the **channel bandwidth B is 5 kHz**.

Likewise, **each signal also has a bandwidth** that measures the maximum range of its frequency components.

- The **faster a signal changes**, the **higher its maximum frequency** is, and the **larger its bandwidth** is.
- **Signals rich in content that changes quickly** (such as those for **battle scenes in a video**) **have larger bandwidth** than **signals that are dull and vary slowly** (such as those for a daytime soap opera or a **video of sleeping animals**).

A signal can be successfully sent over a channel if the **channel bandwidth exceeds the signal bandwidth**. The **rate of information transmission** that a channel can successfully carry is **directly proportional to B** .

- More generally, **if a channel of bandwidth B can transmit N pulses per second, then to transmit KN pulses per second by means of the same technology, we need a channel of bandwidth KB** .
- To repeat, the **number of pulses per second** that can be transmitted over a channel is **directly proportional to its bandwidth B** .

The **signal power P_s** plays a **dual role in information transmission**.

First, P_s is related to the **quality of transmission**. Increasing P_s strengthens the signal pulse and diminishes the effect of channel noise and interference. In fact, **the quality** of either analog or digital **communication systems** **varies with the signal-to-noise ratio (SNR)**.

In any event, **a certain minimum SNR at the receiver is necessary for successful communication**. Thus, a larger signal power P_s allows the system to maintain a minimum SNR over a longer distance, thereby **enabling successful communication over a longer span**.

The **second role of the signal power** is less obvious, although equally important. From the **information theory** point of view, the **channel bandwidth B and the signal power P_s are, to some extent, exchangeable**; that is, **to maintain a given rate and accuracy of information transmission**, we can **trade P_s for B , and vice versa**. Thus,

- **one may use less B if one is willing to increase P_s ,**
- **or one may reduce P_s if one is given a bigger B .**

In short, the **two primary communication resources** are **bandwidth and transmit power**.

- In a given communication channel, **one resource may be more valuable than the other**, and the communication scheme should be designed accordingly.
- A **typical telephone channel**, for example, has **limited bandwidth (3 kHz)**, but the **power is less restrictive**.

- On the other hand, in space vehicles, **huge bandwidth** is available but **the power is severely limited**.
- Hence, the **communication solutions in the two cases are radically different**.

1.3.2 Channel Capacity and Data Rate

Channel bandwidth limits the bandwidth of signals that can successfully pass through, whereas **signal SNR at the receiver determines the recoverability of the transmitted**

- **Higher SNR means** that the **transmitted signal pulse can use more signal levels**, thereby **carrying more bits with** each pulse transmission.
- **Higher bandwidth B also means** that **one can transmit more pulses (faster variation) over the channel**.
- Hence, **SNR and bandwidth B can both affect** the **underlying channel “throughput.”**

The **peak throughput** that can be reliably carried by a channel is defined as the **channel capacity**.

One of the **most commonly encountered channels** is known as the **additive white Gaussian noise (AWGN) channel**.

- The AWGN channel model assumes **no channel distortions** except for the **additive white Gaussian noise and its finite bandwidth B** .
- This **ideal model** captures application cases **with distortionless channels** and
- provides **a performance upper bound** for more general distortive channels.

The **band-limited AWGN channel capacity** was dramatically highlighted by **Shannon's equation**,

$$C = B \log_2 (1 + \text{SNR}) \text{ bit/s} \quad (1.1)$$

Here channel capacity C is the **upper bound of the rate of information transmission per second**. In other words, C is the **maximum number of bits that can be transmitted per second with a probability of error arbitrarily close to zero**; that is, the transmission is as accurate as one desires.

The capacity **only points out this possibility**, however; **it does not specify how it is to be realized**. Moreover, **it is impossible to transmit at a rate higher than this** without incurring errors.

Shannon's equation **clearly brings out the limitation on the rate of communication imposed by B and SNR**.

Practical systems operate at rates below the Shannon rate.

In conclusion, **Shannon's capacity equation demonstrates qualitatively the basic role played by B and SNR in limiting the performance of a communication system**. These

two parameters then represent the ultimate limitation on the rate of communication.

The possibility of resource exchange between these two basic parameters is also demonstrated by the Shannon equation.

As a practical example of trading SNR for bandwidth B , note, however, that this is not a one-to-one trade. Doubling the speaker's volume allows the speaker to talk a little faster, but not twice as fast. This unequal trade effect is fully captured by Shannon's equation [Eq. (1.1)], where doubling the SNR cannot always compensate for the loss of B by 50%.

1.4 MODULATION AND DETECTION

Analog signals generated by the message sources or digital signals generated through A/D conversion of analog signals are often referred to as baseband signals because they typically are low pass in nature.

Baseband signals may be directly transmitted over a suitable channel (e.g., telephone, fax). However, depending on the channel and signal frequency domain characteristics, baseband signals produced by various information sources are not always suitable for direct transmission over a given channel.

- When signal and channel frequency bands do not match exactly, channels cannot be moved.
- Hence, messages must be moved to the right channel frequency bandwidth.
- Message signals must therefore be further modified to facilitate transmission.
- In this conversion process, known as modulation, the baseband signal is used to modify (i.e., modulate), some parameters of a radio-frequency (RF) carrier signal.
 - A carrier is a sinusoid of high frequency. Through modulation, one of the carrier sinusoidal parameters-such as amplitude, frequency, or phase-is varied in proportion to the baseband signal $m(t)$.
 - Accordingly, we have amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM).
 - Figure 1.6 shows a baseband signal $m(t)$ and the corresponding AM and FM waveforms.
 - In AM, the carrier amplitude varies in proportion to $m(t)$, and in FM, the carrier frequency varies in proportion $m(t)$.
 - To reconstruct the baseband signal at the receiver, the modulated signal must pass through a reversal process called demodulation.

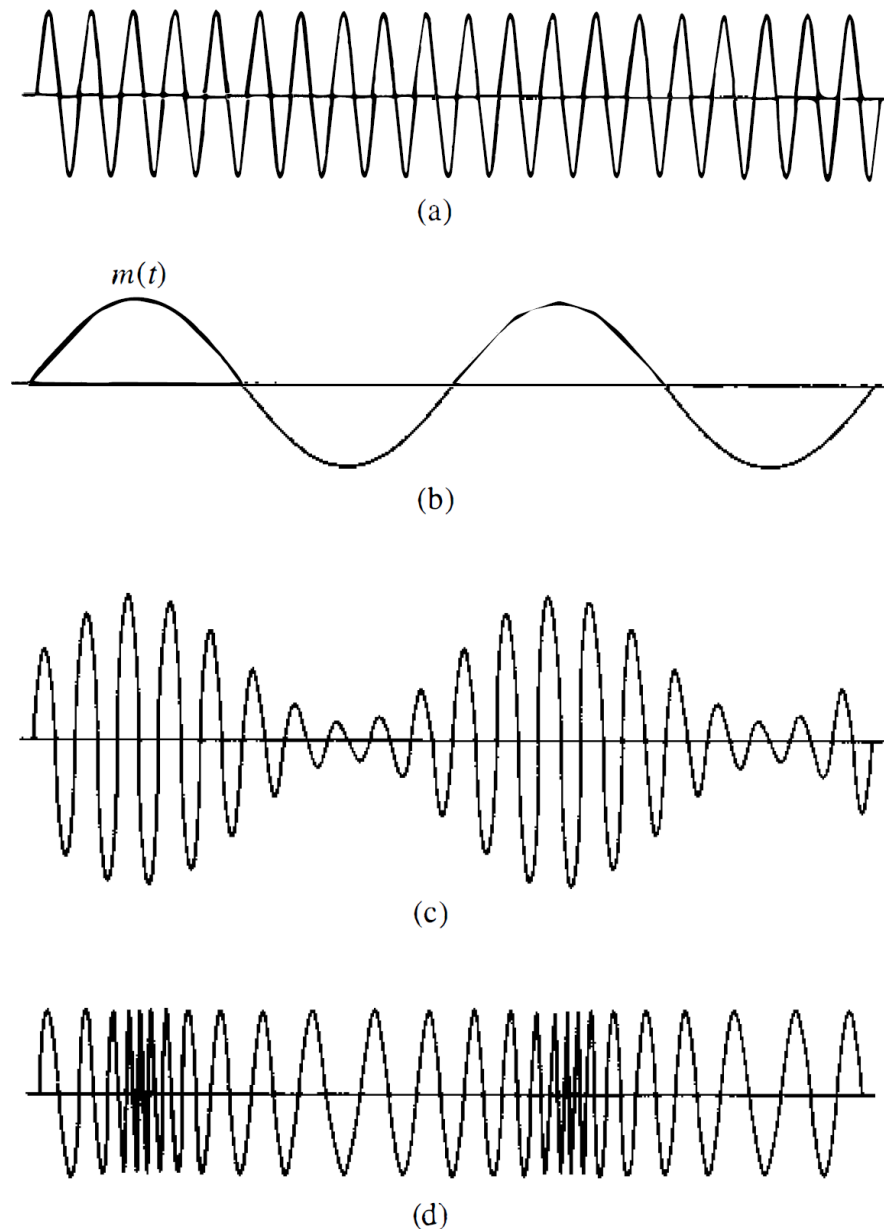


Figure 1.6 Modulation: (a) carrier; (b) modulating (baseband) signal; (c) amplitude modulated wave; (d) frequency modulated wave.

As mentioned earlier, **modulation is used to facilitate transmission**. Some of the **important reasons for modulation** are given next.

1.4.1 Ease of Radiation/Transmission

For efficient radiation of electromagnetic energy, the **radiating antenna should be on the order of a fraction or more of the wavelength of the driving signal.**

For **many baseband signals**, the **wavelengths are too large** for reasonable antenna dimensions. For example,

- the **power in a speech signal** is concentrated at **frequencies** in the range of **100 to 3000 Hz**.
- The corresponding **wavelength is 100 to 3000 km**.
- This long wavelength would **necessitate an impractically large antenna**.

Instead, by **modulating a high-frequency carrier**, we effectively **translate the signal spectrum to the neighborhood of the carrier frequency** that **corresponds to a much smaller wavelength**. For example, a **10 MHz carrier** has a wavelength of only **30 m**, and its transmission can be achieved **with an antenna size on the order of 3 m**.

In this respect, **modulation is** like letting the **baseband signal hitch a ride on a high-frequency sinusoid** (carrier).

The **carrier and the baseband signal may also be compared** to a stone and a piece of paper. If we wish **to throw a piece of paper**, it cannot go too far by itself. But **if it is wrapped around a stone** (a carrier), it can be **thrown over a longer distance**.

1.4.2 Simultaneous Transmission of Multiple Signals- Multiplexing

Modulation also allows multiple signals to be transmitted at the same time in the same geographical area without direct mutual interference.

This case in point is simply demonstrated by considering the **output of multiple television stations carried by the same cable (or over the air) to people's television receivers**.

Without modulation, multiple video signals will all be interfering with one another because **all baseband video signals effectively have the same bandwidth**.

Thus, **cable TV or broadcast TV** without modulation would be **limited to one station at a time** in a given location-a **highly wasteful protocol** *because the channel bandwidth is many times larger than that of the signal*.

One way to solve this problem is to use modulation.

- We can **use various TV stations to modulate different carrier frequencies**, thus translating each signal to a different frequency range.
- If the **various carriers are chosen sufficiently far apart in frequency**, the spectra of the modulated signals (known as TV channels) **will not overlap and thus will not interfere with each other**.
- **At the receiver (TV set), a tunable bandpass filter can select the desired station or TV channel** for viewing.

- This method of transmitting several signals simultaneously, over nonoverlapping frequency bands, is known as **frequency division multiplexing (FDM)**.

Another method of multiplexing several signals is known as **time division multiplexing (TDM)**.

- This method is **suitable when a signal is in the form of a pulse train (as in PCM)**.
- When the pulses are made narrower, the **spaces left between pulses of one user signal are used for pulses from other signals**.
- Thus, in effect, **the transmission time is shared by a number of signals** by interleaving the pulse trains of various signals in a specified order.
- At the receiver, **the pulse trains corresponding to various signals are separated**.

1.4.3 Demodulation

Once multiple modulated signals have arrived **at the receiver**, **the desired signal must be detected and recovered into its original baseband form**.

- Note that **because of FDM**, the first stage of a demodulator typically requires a **tunable bandpass filter** so that the **receiver can select the modulated signal** at a **predetermined frequency band** specified by the transmission station or channel.
- Once a particular modulated signal has been isolated, the demodulator will then need to **convert the carrier variation of amplitude, frequency, or phase, back into the baseband signal voltage**.
- For the **three basic modulation schemes of AM, FM, and PM**, the corresponding demodulators must be designed such that **the detector output voltage varies in proportion to the input modulated signal's amplitude, frequency, and phase, respectively**.
- Once circuits with such response characteristics have been implemented, **the demodulators can downconvert the modulated (RF) signals back into the baseband signals** that **represent the original source message, be it audio, video, or data**.

References

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